

# New Entropy Measures for Tries with Applications to the XBWT

11th Workshop on Data Structures in Bioinformatics

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# What's the Worst-Case Entropy?

## Definition: Worst-Case Entropy

Let  $\mathcal{U}$  be a set, the **worst-case entropy**  $\mathcal{H}^{wc}(\mathcal{U})$  of  $\mathcal{U}$  is defined as

$$\mathcal{H}^{wc}(\mathcal{U}) = \log_2 |\mathcal{U}|$$

Example, if  $\mathcal{U} = \{\text{dog, cat, bird, mouse}\}$ , then  $\mathcal{H}^{wc}(\mathcal{U}) = \log_2 |\mathcal{U}| = \log_2 4 = 2$ .

# What's the Worst-Case Entropy?

If **we assign a unique codeword to every element** of  $\mathcal{U}$ , then there exists an element of  $\mathcal{U}$  having **codeword length** at least  $\mathcal{H}^{wc}(\mathcal{U})$ .

- dog  $\rightarrow$  0
- cat  $\rightarrow$  1
- bird  $\rightarrow$  01
- mouse  $\rightarrow$  11

*Both 'bird' and  
'mouse' have a  
codeword of length  
 $2 = \mathcal{H}^{wc}(\mathcal{U})$*

Let us see how this applies to the case of strings!

# What's the Worst-Case Entropy?

Consider the string **S** = **aaaaaabaaaaaaaaabaaa**.

- If  $\mathcal{U}$  is **the set of strings** of **length**  $n = 20$  and **alphabet**  $\sigma = 2$ :

$$\mathcal{H}^{wc}(\mathcal{U}) = n \log \sigma = 20 \text{ bits}$$

- If  $\mathcal{U}$  is the set of strings where **a** and **b** appear **18** and **2** times:

$$\mathcal{H}^{wc}(\mathcal{U}) = \log \binom{20}{2} \approx 7.57 \text{ bits}$$

Worst-case entropy with fixed frequencies can be much smaller!

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# Worst-Case Entropy vs Empirical Entropy

For a string  $S$ :

- $n_w = \#$  of characters having context  $w \in \Sigma^*$ .
- $n_c = \#$  of characters equal to  $c \in \Sigma$ .

**Worst-case entropy:**  $\mathcal{H}^{wc}(S) = \binom{n}{n_{c_1} \dots n_{c_\sigma}}$

**( $k$ -th) Empirical entropy:**  $\mathcal{H}_k(S) = \sum_{w \in \Sigma^k} \sum_{c \in \Sigma} n_{w,c} \log \frac{n_w}{n_{w,c}}$

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# Our contributions

## ① Extend $\mathcal{H}_k$ and $\mathcal{H}^{wc}$ **from strings to tries!**

Well-known **worst-case trie entropy**  $\log \frac{1}{n} \binom{n\sigma}{n-1}$  [1] without fixed frequencies.

## ② Reachability of our empirical entropy with **arithmetic coding**.

## ③ Comparison between these entropies and other trie measures.

## ④ BWT of a trie can be **compressed** and **indexed** in $\mathcal{H}_k(\mathcal{T}) + o(n)$ .

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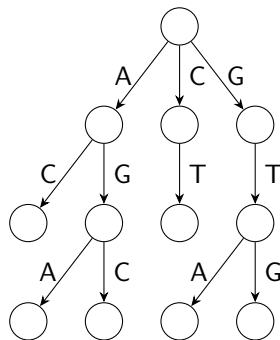
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# Applications

Tries are used in many applications, including: **databases**, **search engines**, **semantic web**, and **NLP**.

In **bioinformatics** they can be used to represent/index genomes or  $k$ -mers [2,3].

Compression achieved only if they share long prefixes.



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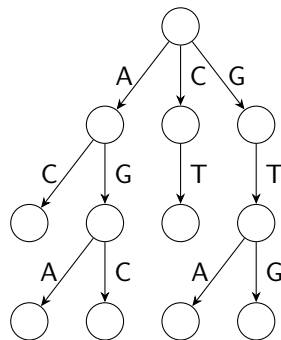
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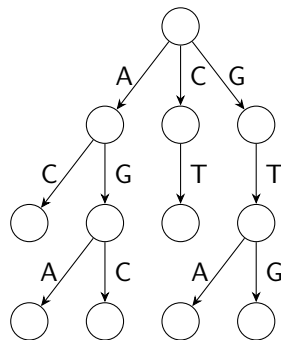
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# Applications

**BWT-based indexes** found many applications in bioinformatics. Ex. are the **FM-index** [4] and the **r-index** [5].

Our index is a **generalisation of the FM-index to tries**, since it compresses to the empirical entropy.



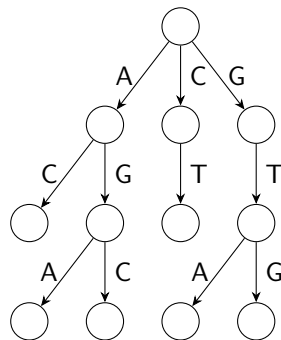
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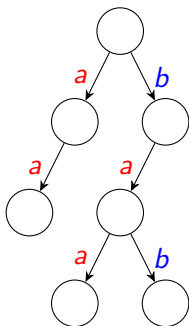
## Worst-Case Entropy with Fixed Frequencies

We consider a **fixed distribution of edge-labels**.

**Example:** trie has 4 edges labeled by *a* and 2 edges labeled by *b*

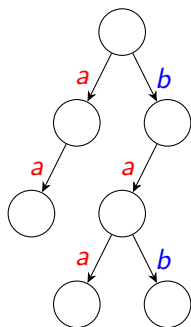
Number of tries  $|\mathcal{U}|$  with a given symbol distribution is:

$$|\mathcal{U}| = \frac{1}{n} \prod_{c \in \Sigma} \binom{n}{n_c}$$

$$n \leftarrow \# \text{ of nodes, } n_c \leftarrow \# \text{ of edges labeled by } c$$




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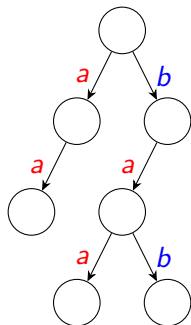
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$n \leftarrow \# \text{ of nodes}$ ,  $n_c \leftarrow \# \text{ of edges labeled by } c$

**Example:** the tries with 7 nodes, 4 edges labeled by **a** and 2 edges labeled by **b** are  $\frac{1}{7} \binom{7}{4} \binom{7}{2} = 105$ .

# Worst-Case Entropy with Fixed Frequencies



Therefore, the **worst-case number of bits** to encode a trie with a fixed symbol distribution is:

$$\mathcal{H}^{wc}(\mathcal{U}) = \log|\mathcal{U}| = \sum_{c \in \Sigma} \log \binom{n}{n_c} - \log n$$

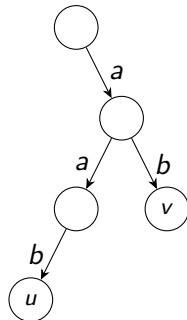
**Example:** To encode a tries with 7 nodes, 4 edges labeled by **a** and 2 edges labeled by **b** we need  $\log \binom{7}{4} + \log \binom{7}{2} - \log 7 \approx 7$  bits in the worst-case!

# Empirical Entropy for Tries

We also extended the **empirical entropy to tries**, our measure:

- 1 Encodes the **labels and the topology simultaneously**.
- 2 Use information on the **edge-labels distribution**.
- 3 Exploits the **k-length context** of the nodes to achieve compression.

**Example:** In figure, **nodes u** and **v** have **2-length context** equal to **ab**.

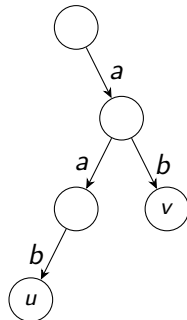


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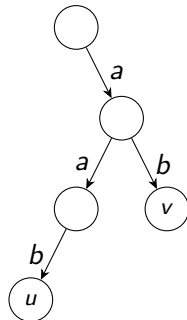


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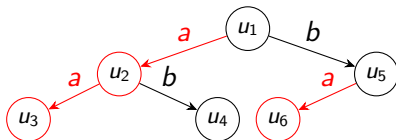
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# Formula Empirical Entropy for Tries

For  $w \in \Sigma^k$  and  $c \in \Sigma$ , consider the integers  $n_w$  and  $n_{w,c}$ :

- $n_w = |\{u \in V \mid u \text{ has context } w\}|$
- $n_{w,c} = |\{u \in V \mid u \text{ has context } w \text{ and there exists } u \xrightarrow{c} v\}|$



**Example:** In figure,  $n_a = 3$ .

Indeed,  $u_2$ ,  $u_3$ , and  $u_6$  are reached by the **string a**.

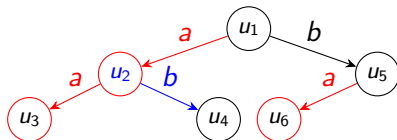
Definition:  $k$ -th order empirical entropy  $\mathcal{H}_k(\mathcal{T})$

$$\mathcal{H}_k(\mathcal{T}) = \sum_{c \in \Sigma} \sum_{w \in \Sigma^k} n_{w,c} \log \left( \frac{n_w}{n_{w,c}} \right) + (n_w - n_{w,c}) \log \left( \frac{n_w}{n_w - n_{w,c}} \right)$$

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**Example:** In figure,  $n_{a,b} = 1$ .

Among the **nodes reached by a**, only  $u_2$  has an **outgoing edge labeled by b**.

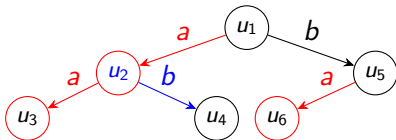
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**Definition: k-th order empirical entropy  $\mathcal{H}_k(\mathcal{T})$**

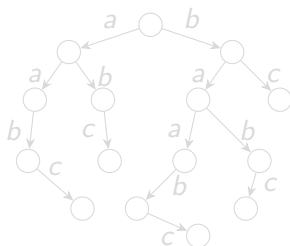
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# Properties for our Entropy Measures

Properties analogous to the string entropies:

- 1  $\mathcal{H}_0(\mathcal{T}) = \mathcal{H}^{wc}(\mathcal{T}) + O(\sigma \log n)$
- 2  $\mathcal{H}_{k+1}(\mathcal{T}) \leq \mathcal{H}_k(\mathcal{T})$ , for every  $k \geq 0$



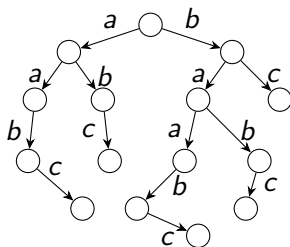
- Worst-case entropy without character frequencies [1] (**not ours!**):  
 $\log \frac{1}{n} \binom{n\sigma}{n-1} = \log \frac{1}{15} \binom{45}{14} \approx \mathbf{33.37 \text{ bits}}$ .
- 1st-order empirical entropy (**ours!**)  
 $n\mathcal{H}_1(\mathcal{T}) \approx \mathbf{7.29 \text{ bits}}$

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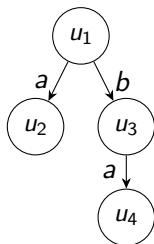
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# Reachability

## Theorem

Every trie  $\mathcal{T}$  can be stored in  $\mathcal{H}_k(\mathcal{T}) + (\sigma + 1)\sigma^k \log n$  bits



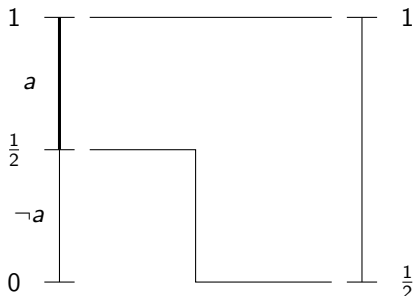
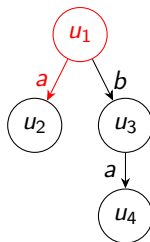
This can be done by extending **arithmetic coding** to tries!

Let see how to compress to the 0-th order  $\mathcal{H}_0(\mathcal{T})$

- Save n. of labels for each character, ex.  $n_a = 2$ ,  $n_b = 1$
- Update a subinterval  $[l, l + s)$  of  $[0, 1)$
- Store a number in  $[l, l + s)$  to reconstruct  $\mathcal{T}$

# Reachability

We iterate over the nodes based on a **preorder visit**:  $u_1, u_2, u_3, u_4$

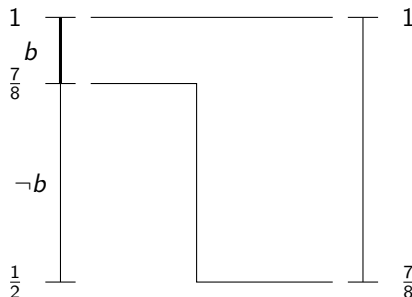
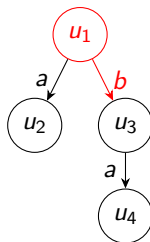


Initial interval  $[0, 1)$ , **probability of outgoing edge labeled by a**:  $\frac{n_a}{n} = \frac{1}{2}$ .

the new interval becomes  $[\frac{1}{2}, 1)$ .

# Reachability

Before moving to  $u_2$ , we redo this operation for the **character b**

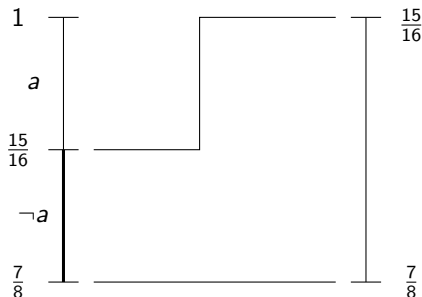
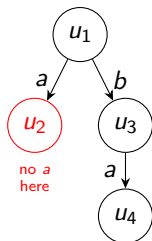


**Probability of outgoing edge labeled by b:**  $\frac{n_b}{n} = \frac{1}{4}$ .

Size of the interval shrinks to  $\frac{1}{4}$ , the new interval becomes  $[\frac{7}{8}, 1)$ .

# Reachability

We move to the **next node in preorder**, i.e.,  $u_2$

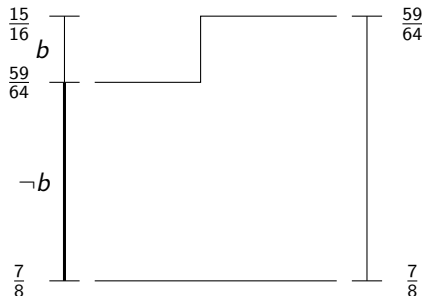
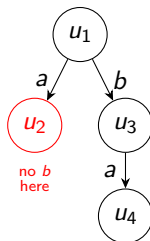


**Probability of NO outgoing edge labeled by  $a$ :**  $1 - \frac{n_a}{n} = \frac{1}{2}$ .

Halve the interval size, the new interval becomes  $[\frac{7}{8}, \frac{15}{16})$ .

# Reachability

And again we redo the same operation for the **character b**

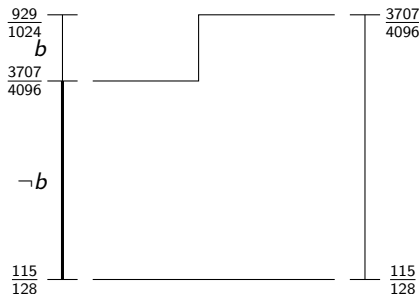
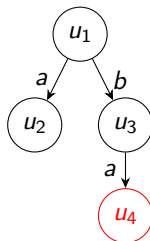


**Probability of NO outgoing edge labeled by b:**  $1 - \frac{n_b}{n} = \frac{3}{4}$ .

The interval size decreases by  $\frac{3}{4}$ , so it becomes  $[\frac{7}{8}, \frac{15}{16})$ .

# Reachability

We continue in this way **until the last node**



The final interval is  $[\frac{115}{2^7}, \frac{3707}{2^{12}})$

We can **store a point in this interval** in at most  $\mathcal{H}_0(\mathcal{T}) + 2$  bits.

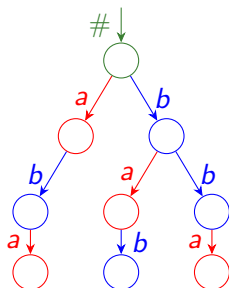


# Comparison with the label entropy

Definition: Label entropy  $\mathcal{H}_k^{label}$  [6]

$cover(w) = \text{labels outgoing from nodes having context } w$

$$\mathcal{H}_k^{label}(\mathcal{T}) = \sum_{w \in \Sigma^k} |cover(w)| \mathcal{H}_0(cover(w))$$



**Example:**

$$cover(\#) \leftarrow ab \quad \mathcal{H}_0(ab) = 1$$

$$cover(a) \leftarrow bb \quad \mathcal{H}_0(bb) = 0$$

$$cover(b) \leftarrow abaa \quad \mathcal{H}_0(abaa) \approx 0.811$$

$$\mathcal{H}_1^{label}(\mathcal{T}) = 1 * 2 + 0 * 2 + 4 * 0.811 \approx 5.245$$

# Comparison with the label entropy

## Theorem

For every  $k \geq 0$  and trie  $\mathcal{T}$ :

$$\mathcal{H}_k(\mathcal{T}) \leq \mathcal{H}_k^{label}(\mathcal{T}) + 1.443n$$

- $\mathcal{H}_k^{label}(\mathcal{T})$  accounts only for the labels, **not the tree topology!**
- To encode the tree topology we need  **$2n - \Theta(\log n)$  bits** [7]

$\mathcal{H}_k(\mathcal{T})$  **always smaller** than  $\mathcal{H}_k^{label}(\mathcal{T}) + 2n - \Theta(\log n)$

For some family of tries  $\mathcal{H}_k(\mathcal{T}) = 0$  and  $\mathcal{H}_k^{label}(\mathcal{T}) = \Omega(n)$ !

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For some family of tries  $\mathcal{H}_k(\mathcal{T}) = 0$  and  $\mathcal{H}_k^{label}(\mathcal{T}) = \Omega(n)$ !

7. G. Navarro, *Compact Data Structures - A Practical Approach*, Cambridge University Press (2016)

# Comparison with the label entropy

## Theorem

For every  $k \geq 0$  and trie  $\mathcal{T}$ :

$$\mathcal{H}_k(\mathcal{T}) \leq \mathcal{H}_k^{label}(\mathcal{T}) + 1.443n$$

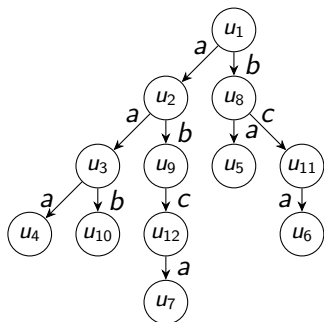
- $\mathcal{H}_k^{label}(\mathcal{T})$  accounts only for the labels, **not the tree topology!**
- To encode the tree topology we need  **$2n - \Theta(\log n)$  bits** [7]

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# XBWT of a trie



$out(u) \leftarrow$  set of outgoing labels of  $u$

$u_1, u_2, \dots, u_n \leftarrow$  nodes sorted **co-lexicographically**

**Definition: XBWT [4]**

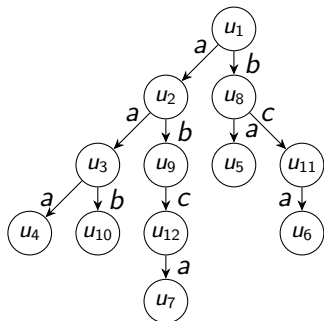
$$XBWT(\mathcal{T}) = out(u_1), out(u_2), \dots, out(u_n)$$

We can **compress** and **index** a trie in:

$$\mathcal{H}_k(\mathcal{T}) + o(n), \forall k = \max\{0, \alpha \log_{\sigma} n - 2\} \text{ s.t. } \alpha < 1$$

| co-lex | $u_1$  | $u_2$  | $u_3$  | $u_4$ | $u_5$ | $u_6$ | $u_7$ | $u_8$      | $u_9$ | $u_{10}$ | $u_{11}$ | $u_{12}$ |
|--------|--------|--------|--------|-------|-------|-------|-------|------------|-------|----------|----------|----------|
| XBWT   | a<br>b | a<br>b | a<br>b |       |       |       |       | a<br><br>c | <br>c |          | a        | a        |

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# Space usage

For the particular case of tries,  $\mathcal{H}_k(\mathcal{T}) + o(n) \leq \mathcal{H}_k^{label}(\mathcal{T}) + 2n + o(n)$ , discussed in the original article about the XBWT [8].

## Theorem Succinctness

If no character appears in more than  $n/2$  edges, then this index is **succinct**, i.e, its space usage is at most:

$$\mathcal{H}^{wc}(\mathcal{T}) + o(\mathcal{H}^{wc}(\mathcal{T}))$$

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8. P. Ferragina et al. Compressing and Indexing Labeled Trees, with Applications. J. ACM (2009)



# Supported operations

These operations are supported **directly on the compressed format**.

|                                                                           | polylog alphabets | arbitrary alphabets                 |
|---------------------------------------------------------------------------|-------------------|-------------------------------------|
| <code>subpath_query(<math>p</math>)</code><br>nodes reached by string $p$ | $O( p )$          | $O( p (\log \sigma + \log \log n))$ |
| <code>parent(<math>u</math>)</code><br>parent node of $u$                 | $O(1)$            | $O(1)$                              |
| <code>c-child(<math>u, c</math>)</code><br>child of $u$ labeled by $c$    | $O(1)$            | $O(1)$                              |
| <code>j-child(<math>u, j</math>)</code><br>$j$ -th child of $u$           | $O(\sigma)$       | $O(\sigma)$                         |

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**This operation is still slow, must be sped up!!!**

Thank you for your attention 😊

- 1 Extend  $\mathcal{H}_k$  and  $\mathcal{H}^{wc}$  **from strings to tries!**
- 2 Reachability of our empirical entropy with **arithmetic coding**.
- 3 Comparison between these entropies and other trie measures.
- 4 BWT of a trie can be **compressed** and **indexed** in  $\mathcal{H}_k(\mathcal{T}) + o(n)$ .

Preprint at: <https://arxiv.org/abs/2512.11618>